

Shifts in microbial thermal traits mitigate heat-induced carbon losses in soils

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Supporting Information Text

Field experiment

Soil samples were collected from the Soil Warming and Nitrogen treatment (SWaN) experiment located on the Prospect Hill Tract of the Harvard Forest Long Term Ecological site in Petersham, Massachusetts, USA (42°50'N, 72°18'W). The forest at the site is composed of mixed hard woods, including red oak (*Quercus rubra*), black oak (*Quercus velutina*), red maple (*Acer rubrum*), striped maple (*Acer pensylvanicum*), American beech (*Fagus grandifolia*), white birch (*Betula papyrifera*), and American chestnut (*Castanea dentata*). The soils are well-drained silty loamy, mixed, mesic, Typic Dystrochrepts (USDA) of the Gloucester series. The mean annual air temperature and precipitation at the site are 7°C and 1100 mm, respectively (Rousk & Frey, 2015). Using a randomized design, 24 3x3 m plots were assigned one of four treatments with six replicates per treatment: 'control', 'nitrogen addition', 'heated', and 'heated x nitrogen addition'. The average soil temperature in warming treatments was continuously elevated to 5°C above ambient using buried heating cables placed at 10 cm depth below the soil surface and spaced 20 cm apart. The cables were installed in 2005, and activated in 2006, providing a period of nine years of warming at the time of sampling. In the period modelled, the mean annual soil temperature in the warmed plots was effectively increased from 9.6°C to 14.2°C (Fig. S3). Nitrogen addition treatments were also initiated in 2006. Nitrogen was applied in equal

doses during the May to October growing season as an aqueous solution of NH_4NO_3 at a rate of 50 kg/ha/y.

Microbial analyses and soil characterization

Microbial respiration was measured as the accumulation of headspace CO_2 in 20 mL glass vials via incubations using a GC-FID (YL6500 GC, YL instruments, Gyeonggi-do, Korea) equipped with a methanizer to measure CO_2 concentrations (as in Rousk & Frey, 2015). Incubation times were also adapted to temperatures, equivalent to ca 14 h at 25°C (i.e., ca. 7 h at 35°C, 28 h at 15°C, and 56 h at 4°C).

Bacterial growth was estimated using leucine (Leu) incorporation in bacteria extracted from soil using the homogenization/centrifugation technique (Bååth, 1994) with modifications (Bååth, Pettersson, & Söderberg, 2001; Rousk & Bååth, 2011). Briefly, 1.0 g of soil was mixed with 20 mL of demineralized water by vortexing (3 min) and centrifuged at low speed (1000 g for 8 min). The resulting bacterial suspension was combined with 2 μL of radio-labelled Leu ($[^3\text{H}]\text{Leu}$, 37 MBq/mL, 5.74 TBq/mmol, Perkin Elmer, UK) and unlabelled Leu, resulting in 275 nM Leu. Incubation times were adapted to incubation temperatures, corresponding to ca 2 h at 25°C (i.e., ca. 45 min at 35°C, 4 h at 16°C, and 8 h at 4°C) to ensure a similar level of carbon and nutrient use in all treatments. Bacterial growth was terminated by adding 75 μL of 100% trichloroacetic acid. After this, samples were washed and the radioactivity of the incorporated Leu was measured on a liquid scintillator. The amount of Leu extracted was finally transformed to units of microbial-carbon (Soares & Rousk, 2019) to obtain the bacterial C-production (i.e., bacterial growth).

Fungal growth was assessed using the acetate (Ac) incorporation into ergosterol method (Newell & Fallon, 1991) adapted for soil (Rousk & Bååth, 2011; Rousk, Brookes, & Bååth, 2009). Briefly, 1.0 g fresh soil was mixed with 20 μL of 1- $[^{14}\text{C}]\text{acetic acid}$ (sodium salt, 7.4 MBq/mL, 2.04 GBq/mmol, Perkin Elmer, UK) and unlabelled sodium acetate resulting in a final acetate concentration of 220 μM in the soil slurry. Samples were then incubated without light, with incubation times adapted to temperatures, corresponding to ca 4 h at 25°C (i.e., ca. 2 h at 35°C, 8 h at 16°C, and 16 h at 4°C). Ergosterol was then extracted, separated and quantified using HPLC equipped with a UV detector. The effluent containing the ergosterol peak was collected and the amount of incorporated radioactivity determined. The amount of Ac incorporated into fungal ergosterol was transformed to units of microbial-carbon (Soares & Rousk, 2019) to obtain the fungal carbon-production (i.e., fungal growth).

Total carbon and nitrogen were estimated using dry dumas combustion, soil pH was measured with water extractions (5:25), water content was determined gravimetrically, and soil density was estimated using sampling rings.

Microbial temperature relationships

To estimate the intrinsic temperature relationship for respiration, bacterial growth and fungal growth, determined at all different screening temperatures, were compared. Then, we modelled the normalized temperature relationships using the Lloyd and Taylor model (Lloyd & Taylor, 1994) for respiration and Ratkowsky model (Ratkowsky, Lowry, McMeekin, Stokes, & Chandler, 1983) for bacterial and fungal growth. These models have been previously used successfully to accurately model microbial relationships across a wide temperature range (e.g., Cruz-Paredes, Tájmél, & Rousk, 2023; Donhauser, Niklaus, Rousk, Larose, & Frey, 2020; Portner, Bugmann, & Wolf, 2010). The Ratkowsky model was additionally used to quantify adaptations to temperature for both respiration and growth by means of the indexes of minimum temperature (T_{min}) and relative temperature sensitivity (Q_{10}) (e.g., Birgander, Olsson, & Rousk, 2018; Cruz-Paredes et al., 2023; Nottingham et al., 2022). The T_{min} index is used to screen for shifts in the temperature acclimation, with higher values showing adaptation to warmer temperatures, and *vice versa* (Donhauser et al., 2020; Nottingham et al., 2022). Q_{10} is used to express the temperature dependence of different biological processes (here defined as the relative difference in rates between 5°C and 15°C), and can be associated with the characteristics of the microbial community composition and the quantity and quality of resources available (Cruz-Paredes, Wallander, Kjølner, & Rousk, 2017; Fierer, Allen, Schimel, & Holden, 2003).

The Lloyd and Taylor model (Lloyd & Taylor, 1994) used to describe respiration relationships was:

$$R = R_r \exp \left(E_0 \left(\frac{1}{T_r - T_0} - \frac{1}{T - T_0} \right) \right) \quad (1),$$

where R represents respiration, R_r is the respiration rates at a reference temperature T_r , E_0 is the activation energy, T_0 is the base temperature, and T is the screening temperature. The model was fit to the data using a single calibration step, where the four parameters were calibrated simultaneously.

The version of the Ratkowsky model used in our study was:

$$\sqrt{R} = a(T - T_{min}) \left(1 - \exp(b(T - T_{max})) \right) \quad (2a),$$

which can be simplified to capture better the behaviour of T_{min} and focus on the relationship below the optimum temperature:

$$\sqrt{R} = a(T - T_{min}) \quad (2b),$$

where R represents the microbial process rate (bacterial growth, fungal growth or respiration), a is a slope parameter related to the absolute rates, b is a slope parameter related to the rate of decline above the optimum temperature, T_{min} (low-temperature x-axis intercept) is the apparent temperature minimum, and T_{max} is the temperature maximum (high-temperature x-axis intercept) (Ratkowsky et al., 1983). The model was applied to the data in two steps as described in Ratkowsky et al. (1983) and Rinnan et al. (2009). First, Eq. 2b was used for screening temperature up to optimum (including temperatures 4°C, 16°C, and 25°C) to estimate the parameters T_{min} and the slope a . In a second step, we used these fitted values for a and T_{min} in Eq. 2a, as constants, and modelled the temperature relationship between T_{min} to T_{max} using data from all screening temperatures.

To estimate Q_{10} , indicating temperature sensitivity, the following equation was used:

$$Q_{10} = \frac{(T+10-T_{min})^2}{(T-T_{min})^2} \quad (3),$$

where Q_{10} describes how the rate changes with a temperature change of 10°C and was calculated for the temperature range 5-15°C.

Microbial temperature relationships were differentiated for bacteria and fungi, as their rates were better described by the Ratkowsky model when fitted separately. However, when assessing the *in situ* rates and effects on SOC stocks, the contributions of bacteria and fungi were pooled together.

Effects of soil moisture on carbon fluxes and SOC stocks

To model *in situ* microbial process rates considering the effects of soil moisture, we utilised soil moisture data continuously recorded from other plots at the same experimental site (<10 m from the SWaN plots) at hourly resolution (using 2016 as a proxy for a full annual cycle) (Fig. S3A) and the modelled process rates over the year (excluding the effect of water). The *in situ* rates could then be calculated (as in Sierra, Trumbore, Davidson, Vicca, & Janssens, 2015), considering temperature (T) and moisture (θ) as independent (Cruz-Paredes, Tájmel, & Rousk, 2021) abiotic factors influencing decomposition rates:

$$r(t) = r_T(T(t)) f(\theta(t)) \quad (4),$$

where r_T is the *in situ* rates obtained by the scenario considering changes in temperature only, and $f(\theta)$ denoted the reduction function due to soil moisture limitation based on the CENTURY model (based on Bauer, Herbst, Huisman, Weihermüller, & Vereecken, 2008; Parton et al., 1993):

$$f(\theta) = \frac{1}{1+30\exp(-8.5\theta/\theta_h)} \quad (5),$$

where θ_h is the volumetric water content at 0.1 MPa, calibrated for the silty loam soil under study (Carsel & Parrish, 1988; van Genuchten, 1980). However, it is important to acknowledge that this approach has limitations. It shared with most moisture rate modifiers used in soil decomposition models (e.g., Roth-C, Daycent, Orchidee) that it assumes identical moisture dependencies for all microbial process rates (thus not distinguishing respiration, bacterial growth, and fungal growth), and that rate modifiers do not differ between different environments.

Phospholipid fatty acids (PLFAs)

PLFAs were extracted from soil samples following the method described in (Frostegård, Bååth, & Tunlio, 1993). In brief, soil was extracted using a single-phase chloroform:methanol:citrate buffer mixture (1:2:0.8 v/v/v), and phospholipids were isolated using a silicic acid column. The phospholipids were then analyzed on a gas chromatograph with a flame ionization detector. Methyl nonadecanoate fatty acid (19:0) was added as an internal standard prior to methylation for quantification of peak areas. The relative abundances of the 32 most abundant PLFAs (i14:0, 14:0, i15:0, a15:0, 15:0, br16:0, i16:0, 16:1 ω 9, 16:1 ω 7c, 16:1 ω 7t, 16:1 ω 5, 16:0, br17:0, 10Me16:0a, 10Me16:0b, i17:0, a17:0, 17:1 ω 8, cy17:0, 17:0, br18:0, 10Me17:0, x3, 18:2 ω 6, 18:1 ω 9, 18:1 ω 7, 18:1, 18:0, 19:1a, 10Me18:0, cy19:0, 20:0) were analyzed using principal component analysis (PCA), with data standardized to unit variance.

Supporting Figures and Tables

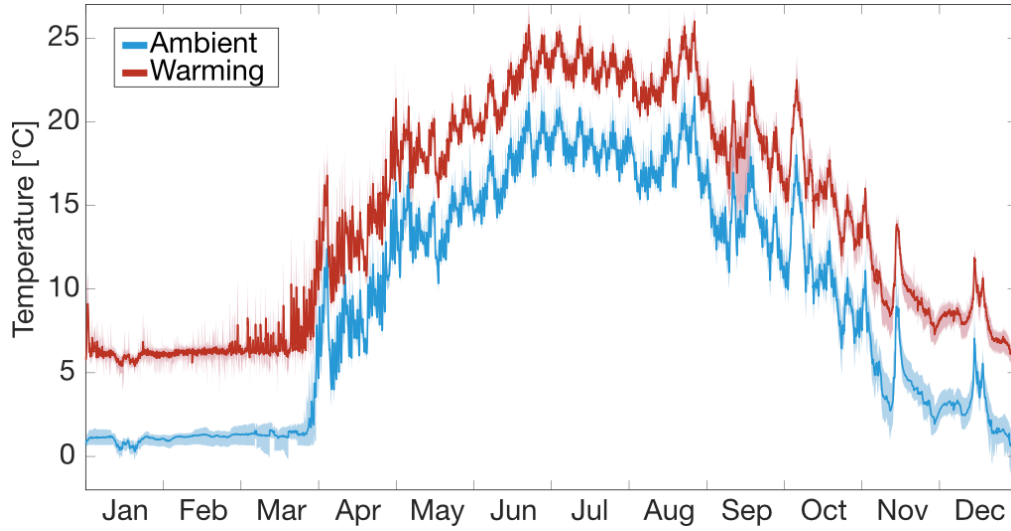


Fig. S1. Soil temperature recorded at the field site, illustrating the year-round difference of +5°C between the ambient and warming plots.

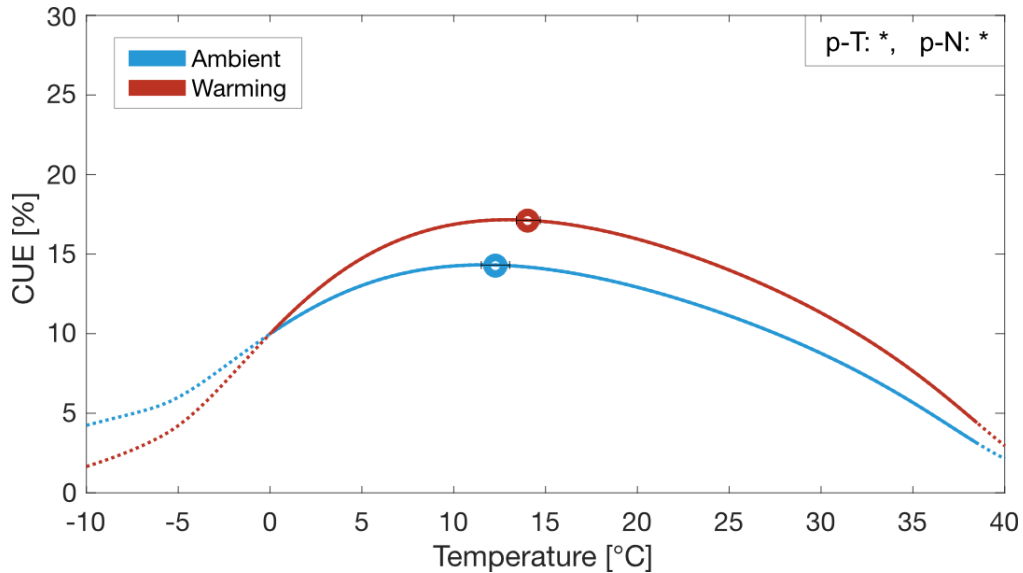


Fig. S2. Carbon use efficiency across temperature for communities under ‘ambient’ and ‘warming’ treatments. Markers show the temperatures of maximum CUE, with error bars as their standard errors. Dotted lines mark estimations based on extrapolations of laboratory data ($\pm 4^\circ\text{C}$ of measured range). The p-values indicate the significance of the differences between temperature (p-T) and nitrogen-addition treatments (p-N). Significance level: *, $0.01 < p < 0.05$. Refer to Table S4 for detailed statistics.

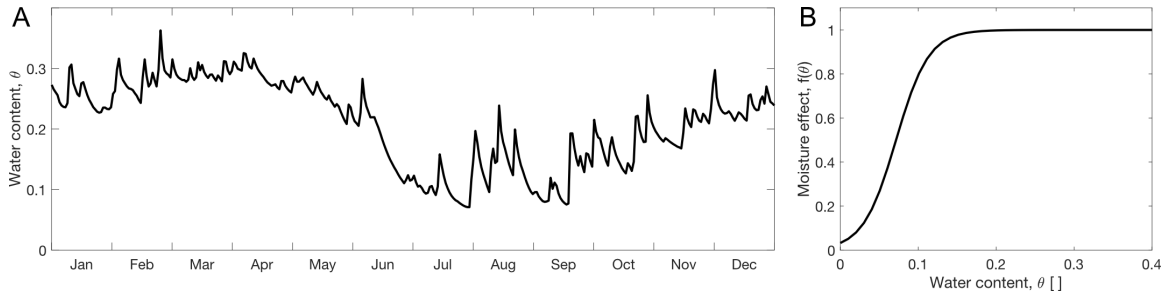


Fig. S3. Soil moisture recorded at the field site (volumetric water content; θ) (panel A) and moisture reduction function used ($f(\theta)$) (panel B).

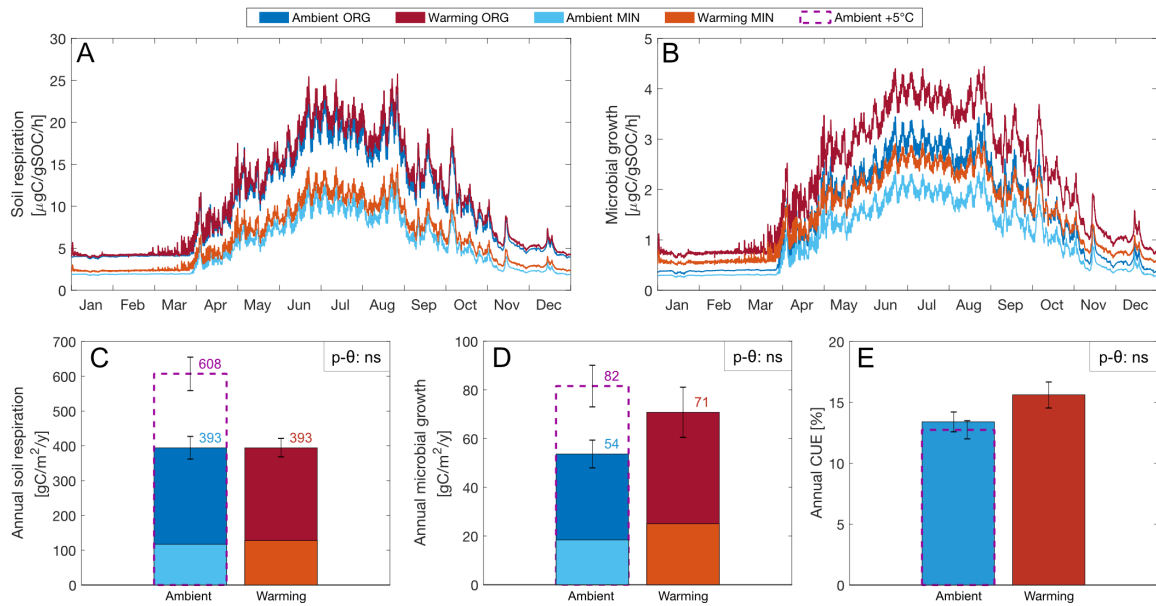


Fig. S4. Modelled rates without considering limitations by water availability. Panels show microbial growth (A) and respiration (B) throughout a natural year, and annual microbial growth (C), microbial CO_2 emissions (D) and C-use efficiency (E). The p-values indicate the significance of the differences between scenarios with and without considering moisture limitations ($p > 0.05$; refer to Table S7 for detailed statistics).

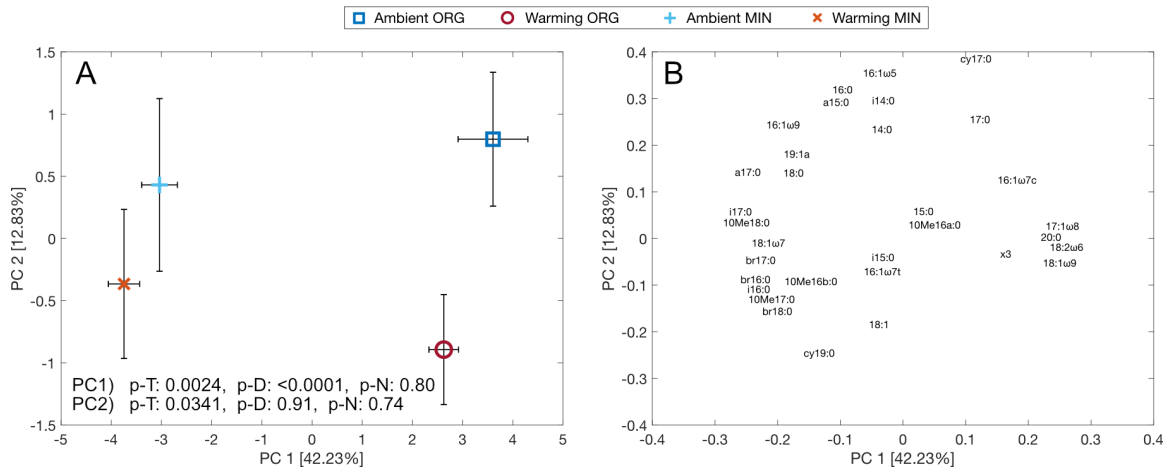


Fig. S5. Principal Component Analysis (PCA) of the phospholipid fatty acid (PLFA) composition in soil samples. Panel A shows the average scores for each warming treatment and depth for principal components 1 (PC1) and 2 (PC2), where error bars represent the standard error. The p-values indicate the significance of the differences between temperature treatments (p-T), soil depths (p-D), and nitrogen-addition treatments (p-N). Panel B shows the loadings for individual PLFA in the PCA.

Table S1. Detailed statistics of main soil characteristics.

		Total C	Total N	C:N	pH	WC	ρ	SOC
T	p	0.0492	0.16	0.0922	0.22	0.0004	0.12	0.28
	F _{1,33}	4.17	2.09	3.01	1.60	15.73	2.49	1.28
D	p	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	-
	F _{1,33}	343.33	185.73	19.15	190.91	83.77	91.56	-
N	p	0.13	0.31	0.22	0.78	0.55	0.57	0.99
	F _{1,33}	2.38	1.06	1.57	0.08	0.37	0.32	0.00
T x D	p	0.63	0.31	0.24	0.27	0.0483	0.18	-
	F _{1,33}	0.24	1.07	1.42	1.27	4.31	1.85	-
T x N	p	0.34	0.32	0.83	0.19	0.84	0.12	0.80
	F _{1,33}	0.95	1.04	0.04	1.76	0.04	2.49	0.07
D x N	p	0.0069	0.31	0.0165	0.55	0.80	0.42	-
	F _{1,33}	8.29	1.09	6.38	0.37	0.06	0.67	-
T x D x N	p	0.81	0.87	0.40	0.92	0.61	0.86	-
	F _{1,33}	0.06	0.03	0.72	0.01	0.26	0.03	-

Note: Soil characteristics highlighted the distinctive features between organic and mineral horizons. The differences across warming treatments were only statistically significant for total carbon and water content at the time of sampling (WC). Chronic warming reduced SOC by 15% and soil moisture by 39%, on average. Nitrogen addition did not lead to significant differences in any soil characteristics.

Table S2. Detailed statistics of $T_{\min}$ and Q_{10} derived from the temperature sensitivities of soil respiration, bacterial growth, and fungal growth.

		Soil respiration		Bacterial growth		Fungal growth	
		$T_{\min}$ [°C]	Q_{10}	$T_{\min}$ [°C]	Q_{10}	$T_{\min}$ [°C]	Q_{10}
Ambient ORG		-9.8 ± 0.5	2.9 ± 0.1	-7.3 ± 0.2	3.3 ± 0.1	-21.3 ± 1.7	2.0 ± 0.1
Warming ORG		-8.1 ± 0.3	3.1 ± 0.1	-5.0 ± 0.4	4.0 ± 0.2	-14.2 ± 2.2	2.7 ± 0.3
Ambient MIN		-8.7 ± 0.3	3.0 ± 0.1	-8.0 ± 1.6	3.4 ± 0.2	-19.4 ± 1.2	2.0 ± 0.1
Warming MIN		-8.7 ± 0.9	3.1 ± 0.1	-6.6 ± 0.6	3.5 ± 0.2	-13.5 ± 0.9	2.4 ± 0.1
T	p	0.1600	0.1200	0.0395	0.0052	0.0006	0.0040
	$F_{1,33}$	2.04	2.57	4.60	8.94	14.33	9.56
D	P	0.6200	0.5700	0.1835	0.1689	0.4700	0.4800
	$F_{1,33}$	0.25	0.33	1.85	1.98	0.54	0.52
N	p	0.5000	0.7200	0.6100	0.9200	0.1200	0.3300
	$F_{1,33}$	0.47	0.13	0.26	0.01	2.50	0.96
T x D	p	0.1500	0.3300	0.6100	0.05	0.7100	0.3500
	$F_{1,33}$	2.14	0.97	0.27	4.11	0.14	0.89
T x N	p	0.2500	0.2800	0.3200	0.3400	0.3500	0.7600
	$F_{1,33}$	1.36	1.21	0.99	0.94	0.88	0.10
D x N	p	0.6700	0.7500	0.5900	0.9800	0.7000	0.7100
	$F_{1,33}$	0.19	0.10	0.30	0.00	0.15	0.14
T x D x N	p	0.8400	0.9300	0.1800	0.2000	0.9700	0.8800
	$F_{1,33}$	0.04	0.01	1.89	1.74	0.00	0.02

Note: The table presents mean values $\pm$ standard error.

Table S3. Detailed statistics of the rates at 25°C of temperature relationships for soil respiration, bacterial growth, and fungal growth.

		Microbial respiration	Bacterial growth	Fungal growth
T	p	0.0079	0.85	0.0262
	$F_{1,33}$	7.99	0.04	5.42
D	p	<0.0001	0.0016	<0.0001
	$F_{1,33}$	36.94	11.77	63.99
N	p	0.99	0.0547	0.59
	$F_{1,33}$	0.00	3.97	0.30
T x D	p	0.75	0.45	0.46
	$F_{1,33}$	0.11	0.59	0.57
T x N	p	0.50	0.27	0.68
	$F_{1,33}$	0.46	1.23	0.18
D x N	p	0.0037	0.16	0.0743
	$F_{1,33}$	9.78	2.08	3.40
T x D x N	p	0.96	0.42	0.0010
	$F_{1,33}$	0.00	0.66	7.48

Table S4. Detailed statistics of the “optimal” temperature (maximum) of carbon use efficiency.

		T [°C]
Ambient		12.3 ± 0.8
Warming		14.1 ± 0.7
T	p	0.0215
	F _{1,14}	6.69
N	p	0.0359
	F _{1,14}	5.38
T x N	p	0.08
	F _{1,14}	3.47

Table S5. Detailed statistics of modelled annual soil respiration, production of microbial growth, and carbon use efficiency in soils per unit soil surface area.

		Annual soil emissions	Annual microbial growth	CUE
T	p	0.77	0.14	0.0318
	F _{1,14}	0.09	2.43	6.68
N	p	0.48	0.06	0.0187
	F _{1,14}	0.53	4.29	7.07
T x N	p	0.52	0.52	0.61
	F _{1,14}	0.25	0.44	0.28

Table S6. Detailed statistic of the modelled annual soil respiration, microbial growth, and carbon use efficiency to evaluate the direct vs indirect effects of temperature.

		Direct effect: Ambient +5°C vs Ambient			Indirect effect: Ambient +5°C vs Warming		
		Annual soil emissions	Annual microbial growth	CUE	Annual soil emissions	Annual microbial growth	CUE
Eff	p	0.0486	0.0481	0.78	0.0660	0.81	0.0205
	F _{1,16}	3.96	3.83	0.08	3.98	0.06	6.83
N	p	0.27	0.0080	0.0006	0.46	0.0598	0.0165
	F _{1,16}	1.27	9.15	18.29	0.57	4.19	7.41
Eff x N	p	0.99	0.98	0.95	0.61	0.51	0.57
	F _{1,16}	0.00	0.00	0.00	0.27	0.45	0.34

Table S7. Comparison of modelled annual fluxes and carbon use efficiency with and without considering water availability limitations.

		Annual soil emissions	Annual microbial growth	CUE
θ	p	0.56	0.61	0.99
	F _{1,31}	0.35	0.27	0.00

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